Hawkeye EQMS · Design Control (medical device) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

Document version	**v1.0**
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Tenant under test	Novex Pharma Inc. · `69e64e7869b2ba745d40bb89`
Live frontend	`https://hawkeye-frontend-dev-chi.vercel.app`
Live backend	`https://hawkeye-backend-dev.vercel.app`

Contents

1. Module overview

Module	**Design Control (medical device)**
Pharma purpose	Track a medical-device design through 8 phases (Planning → Input → Output → Review → Verification → Validation → Transfer → Changes). Maintain the Design History File (DHF). Link to Risk, Change Control, CAPA.
Compliance basis	21 CFR 820.30 (Design Controls) · ISO 13485:2016 §7.3 · ISO 14971 (risk management) · MDR 2017/745
Hawkeye routes	`/design-controls (frontend)` · `/api/design-controls (backend)`
Hawkeye model file	`backend/src/models/DesignControlModel.js`
Tenant module flag	`modules.DESIGN_CONTROL`

Designs follow DRAFT → ACTIVE → DESIGN_FREEZE → TRANSFERRED → OBSOLETE. The 8 design phases (PLANNING → INPUT → OUTPUT → REVIEW → VERIFICATION → VALIDATION → TRANSFER → CHANGES) progress sequentially via /advance-phase. Each phase status: NOT_STARTED → IN_PROGRESS → COMPLETED → BLOCKED.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
8-phase design lifecycle per 21 CFR 820.30 21 CFR 820.30	phaseKey enum with 8 phases · sequential transition rule.	MET	
Device classification (Class I / II / III / IVD) 21 CFR 860	deviceClass enum · 4 values.	MET	
Regulatory pathway (510K / PMA / De Novo / CE Mark) 21 CFR 807, 814	regulatoryPathway enum · 5 values.	MET	
Design Inputs (Requirements) with traceability to outputs 21 CFR 820.30(c)	designInputs[] subdocument with requirementId + source + priority + verified + traceableToOutput.	MET	
Design Outputs (Specifications) traceable to inputs 21 CFR 820.30(d)	designOutputs[] with outputType + documentRef + tracedToInput.	MET	
Design Reviews with attendees + decision (PROCEED / REVISE / HOLD) 21 CFR 820.30(e)	designReviews[] subdocument.	MET	
Verification (V&V protocols + reports) 21 CFR 820.30(f)	verificationProtocolRef + verificationReportRef fields.	MET	
Validation (V&V protocols + reports) 21 CFR 820.30(g)	validationProtocolRef + validationReportRef fields.	MET	
Design Transfer with manufacturing site link 21 CFR 820.30(h)	POST /:id/transfer with manufacturingSiteId.	MET	
Design History File (DHF) 21 CFR 820.30(j)	dhfRef field linking to controlled-document repository.	MET	
Linked Risk / Change Control / CAPA ISO 14971	linkedRiskItems[] + linkedChangeControlIds[] + linkedCapaIds[].	MET	
Auto-numbered (DC-YYYY-NNNN) GMP traceability	Pre-save generator.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026`. Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()`.

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Product Engineer (custom role) engineer	`(engineer-side login)`	Authors design, captures inputs + outputs, advances phases.	DRAFT ACTIVE
Head of QA James admin · reviewer	`qa.head@novex-pharma.demo`	Reviews phase artifacts, signs off design reviews, approves verification.	ACTIVE (REVIEW + VERIFICATION)
Marcus Brown Reg Affairs	`regulatory@novex-pharma.demo`	Verifies regulatory pathway, approves transfer.	ACTIVE (TRANSFER)
Manufacturing (Production Head Michael) production	`production.head@novex-pharma.demo`	Receives transferred design, accesses DHF.	TRANSFERRED

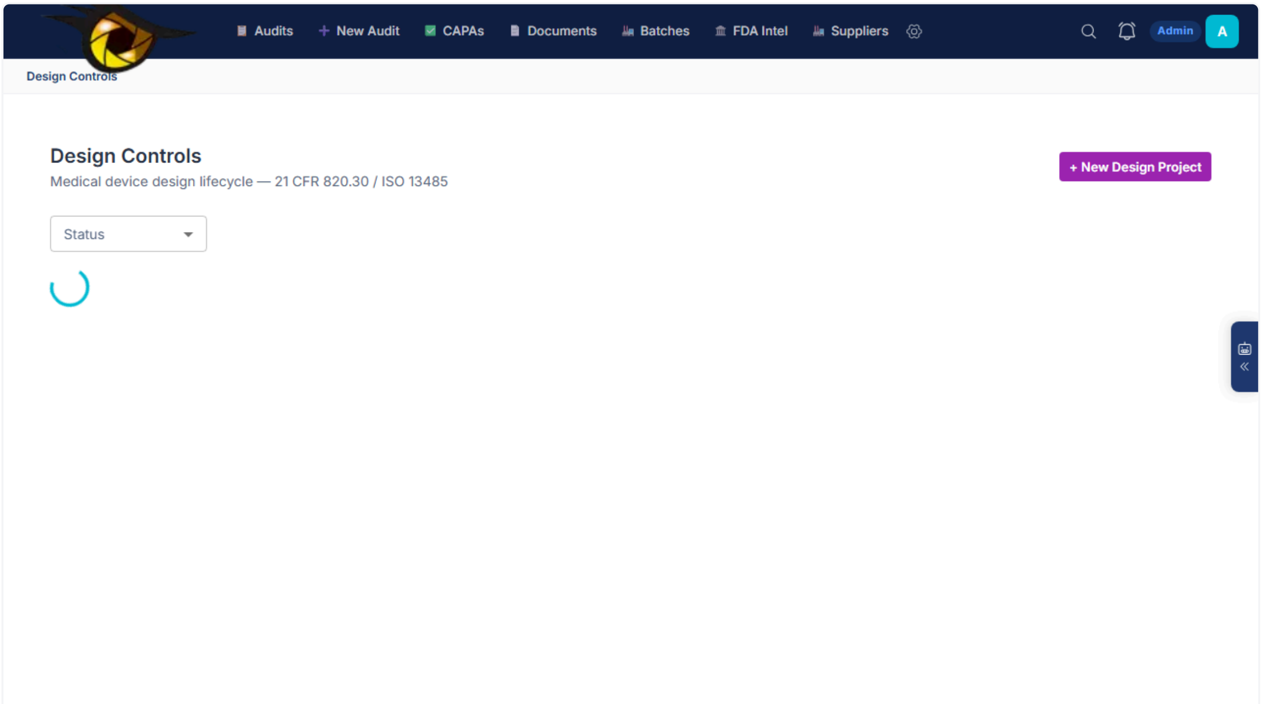
4. Feature catalogue (click-by-click)

4.1 Design register

What it does	Lists all designs with deviceClass + regulatoryPathway + currentPhase + status.
Menu / route	`/design-controls`
Available to roles	any tenant viewer
Backend API	`GET /api/design-controls`

CLICK-BY-CLICK

1. NAVIGATE Click 'Design Controls' (admin nav)
   → Page renders



state-screens/design-control-list.png

4.2 + Start Design dialog

What it does	Author a new design.
Menu / route	`/design-controls · top-right '+ Start Design'`
Available to roles	**engineer · admin**
Backend API	`POST /api/design-controls`

CLICK-BY-CLICK

1. **CLICK** Click '+ Start Design'
   → Dialog opens
2. **TYPE** Fill title + productName
   → required
3. **CLICK** Pick deviceClass (CLASS_I / II / III / IVD)
   → required
4. **CLICK** Pick regulatoryPathway (510K / PMA / DE_NOVO / CE_MARK / OTHER)
   → required
5. **CLICK** Click 'Save'
   → Status=DRAFT · designNumber=DC-YYYY-NNNN · 8 phases initialized to NOT_STARTED

FIELDS CAPTURED

Field	Required	Validation / values	Notes
`title`	Yes	string	
`productName`	Yes	string	

Field	Required	Validation / values	Notes
`deviceClass`	Yes	CLASS_I | CLASS_II | CLASS_III | IVD	
`regulatoryPathway`	Yes	510K | PMA | DE_NOVO | CE_MARK | OTHER	
`riskLevel`	No	LOW | MEDIUM | HIGH | UNACCEPTABLE	

4.3 Advance phase

What it does	Marks current phase COMPLETED + next phase IN_PROGRESS. Auto-flips overall status when last phase done.
Menu / route	`Design detail · 'Advance to next phase' button`
Available to roles	**engineer · QA**
Backend API	`POST /api/design-controls/:id/advance-phase`

CLICK-BY-CLICK

1. **CLICK** Click 'Advance to next phase'
   → Phase N → COMPLETED · Phase N+1 → IN_PROGRESS · status auto-flips DRAFT → ACTIVE on first advance

4.4 Add Design Review

What it does	Capture a design review meeting with attendees + decision (PROCEED / REVISE / HOLD).
Menu / route	`Design detail · 'Add Review' button (visible from REVIEW phase onwards)`
Available to roles	**QA · reviewer**
Backend API	`POST /api/design-controls/:id/reviews`

CLICK-BY-CLICK

1. **CLICK** Click 'Add Review'
   → Drawer opens
2. **CLICK** Pick reviewDate + reviewPhase
   → required
3. **CLICK** Pick attendees[]
   → ObjectId[]
4. **CLICK** Pick decision (PROCEED / REVISE / HOLD)
   → required
5. **TYPE** Fill actionItems[] + minutesRef
6. **CLICK** Click 'Save'
   → designReviews[] += entry

4.5 Add Design Input / Output

What it does	Capture requirements (inputs) + specifications (outputs) with traceability.

Menu / route	`Design detail · 'Inputs' / 'Outputs' tabs`
Available to roles	**engineer**
Backend API	`PUT /api/design-controls/:id (with new entries)`

CLICK-BY-CLICK

1. **CLICK** Open Inputs tab · '+ Add Input'
   → Form opens
2. **TYPE** Fill requirementId + description
   → required
3. **CLICK** Pick source (USER_NEED / REGULATORY / STANDARD / RISK / BUSINESS)
4. **CLICK** Pick priority (CRITICAL / HIGH / MEDIUM / LOW)
5. **CLICK** Save
   → designInputs[] += entry

4.6 Design Transfer (TRANSFER phase)

What it does	Formally transfer design to manufacturing site.
Menu / route	`Design detail · 'Transfer' button (visible in TRANSFER phase)`
Available to roles	**QA · RA · admin**
Backend API	`POST /api/design-controls/:id/transfer`

CLICK-BY-CLICK

1. **CLICK** Click 'Transfer'
   → Drawer opens
2. **CLICK** Pick manufacturingSiteId
   → required
3. **CLICK** Click 'Confirm transfer'
   → Status flips DRAFT/ACTIVE → DESIGN_FREEZE → TRANSFERRED · currentPhase=CHANGES

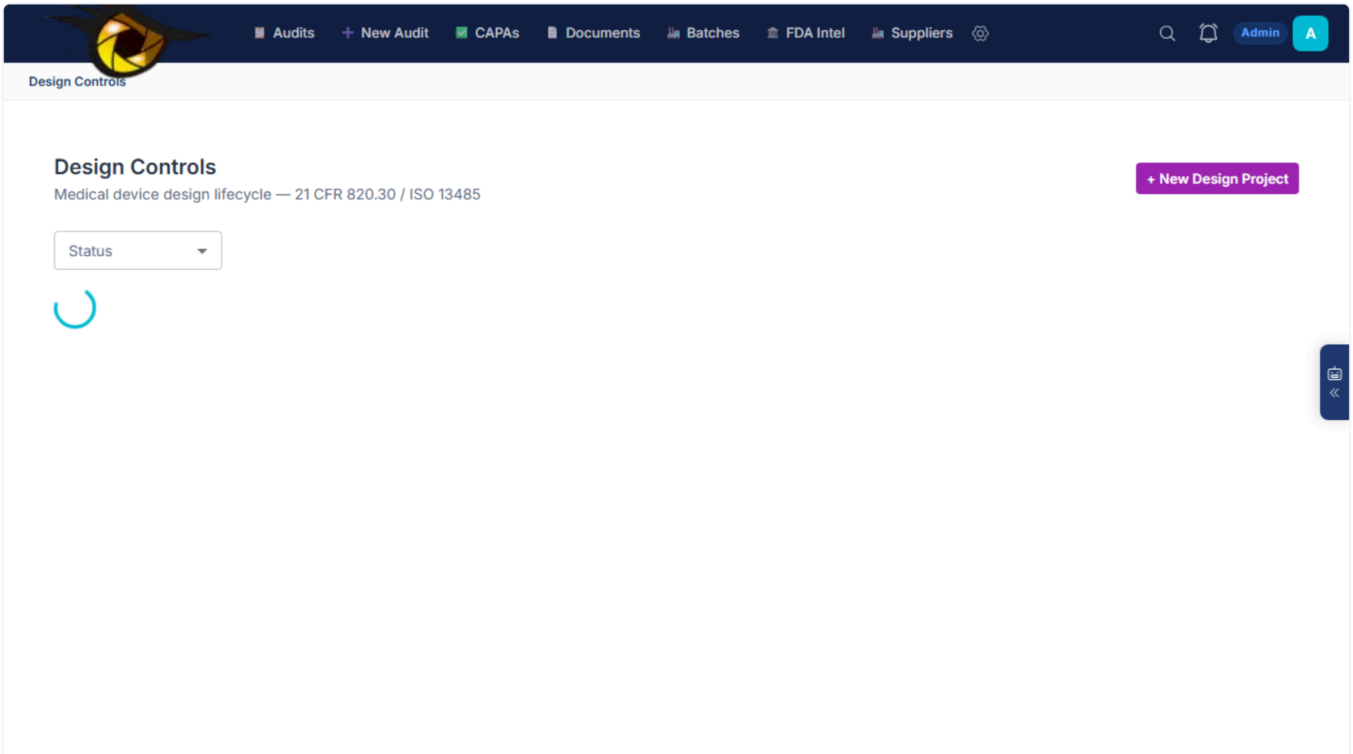
5. Full lifecycle walkthrough — one transaction, end-to-end

One Class II 510(k) device walked from PLANNING through TRANSFER.

#	Persona	State transition	Action	API	Observed	Result
1	**Product Engineer** engineer	`–` → `DRAFT`	+ Start Design · CLASS_II · 510K	`POST /api/design-controls`	designNumber=DC-2026-NNNN · status=DRAFT · 8 phases initialized	**PASS**

Expected DB state after this step:

```
design-controls { _id, designNumber, deviceClass: 'CLASS_II', regulatoryPathway: '510K', currentPhase: 'PLANNING',
phases: [8 entries with status NOT_STARTED], status: 'DRAFT' }
```

<table>
<tr><th>#</th><th>Persona</th><th>State transition</th><th>Action</th><th>API</th><th>Observed</th><th>Result</th></tr>
<tr><td colspan="7">


state-screens/design-control-list.png
</td></tr>
<tr><td>2</td><td>Engineer
engineer</td><td><code>DRAFT</code> → <code>ACTIVE</code></td><td>Advance phase (PLANNING starts) → status auto-flips ACTIVE</td><td><code>POST /api/design-controls/:id/advance-phase</code></td><td>Status=ACTIVE · phases[0].status=IN_PROGRESS</td><td>PASS</td></tr>
<tr><td>3</td><td>Engineer
engineer</td><td><code>ACTIVE (PLANNING)</code> → <code>ACTIVE (INPUT)</code></td><td>Capture design inputs · Advance phase</td><td><code>POST /api/design-controls/:id/advance-phase</code></td><td>phases[0]=COMPLETED · phases[1]=IN_PROGRESS</td><td>PASS</td></tr>
<tr><td>4</td><td>Engineer
engineer</td><td><code>ACTIVE (INPUT)</code> → <code>ACTIVE (OUTPUT)</code></td><td>Add design outputs traceable to inputs · Advance</td><td><code>POST /api/design-controls/:id/advance-phase</code></td><td>phases[1]=COMPLETED · phases[2]=IN_PROGRESS</td><td>PASS</td></tr>
<tr><td>5</td><td>James
QA reviewer</td><td><code>ACTIVE (REVIEW)</code> → <code>ACTIVE (VERIFICATION)</code></td><td>Add design review · decision=PROCEED · Advance</td><td><code>POST /api/design-controls/:id/reviews</code> <code>+ advance-phase</code></td><td>designReviews[] populated · phase advances</td><td>PASS</td></tr>
<tr><td>6</td><td>James
QA</td><td><code>ACTIVE (VERIFICATION)</code> → <code>ACTIVE (VALIDATION)</code></td><td>Attach verificationProtocolRef + verificationReportRef · Advance</td><td><code>PUT + advance-phase</code></td><td>Phase advances</td><td>PASS</td></tr>
</table>

#	Persona	State transition	Action	API	Observed	Result
7	**Marcus** Reg Affairs	`ACTIVE (VALIDATION)` → `ACTIVE (TRANSFER)`	Attach validation report · Advance	`PUT + advance-phase`	Phase advances	PASS
8	**Marcus + James** RA + QA	`ACTIVE (TRANSFER)` → `TRANSFERRED`	POST /transfer · manufacturingSiteId	`POST /api/design-controls/:id/transfer`	Status=TRANSFERRED · currentPhase=CHANGES · DHF locked	PASS

Expected DB state after this step:

```
design-controls { status: 'TRANSFERRED', currentPhase: 'CHANGES', dhfRef }
```

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
(roadmap) AI design-input traceability checker	`ACTIVE (INPUT/OUTPUT)`	`(future)`	Inputs / Outputs tabs	Cross-check that every input has at least one output that traces to it; flag orphans	rule + Free Gemini for narrative

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
`ACTIVE (PLANNING)`	`21 CFR 820.30(b)`	Design plan + responsibilities + reviews schedule	design-controls
`ACTIVE (INPUT)`	`21 CFR 820.30(c)`	designInputs[] with source + priority + verified flag	design-controls
`ACTIVE (OUTPUT)`	`21 CFR 820.30(d)`	designOutputs[] with documentRef + tracedToInput	design-controls
`ACTIVE (REVIEW)`	`21 CFR 820.30(e)`	designReviews[] with attendees + decision + minutesRef	design-controls
`ACTIVE (VERIFICATION)`	`21 CFR 820.30(f)`	verificationProtocolRef + verificationReportRef	design-controls
`ACTIVE (VALIDATION)`	`21 CFR 820.30(g)`	validationProtocolRef + validationReportRef	design-controls

State	Cited clauses	Evidence captured	Backing records
`TRANSFERRED`	`21 CFR 820.30(h)` `21 CFR 820.30(j)`	manufacturingSiteId + dhfRef + transferredAt	design-controls

8. Test results matrix

Suite	Scope	Result	Evidence
(deferred) eqms-lifecycle.spec.ts · design-control	Full 8-phase walkthrough	SKIP	Not yet automated end-to-end

9. Known gaps + roadmap

- **Full 8-phase E2E lifecycle test** — Engineer + QA + RA + Manufacturing orchestration spec. MEDIUM
- **AI traceability checker** — Cross-check inputs ↔ outputs ↔ verification ↔ validation chain. MEDIUM
- **DHF auto-assembler on TRANSFER** — Auto-bundle all phase artifacts into a single PDF DHF on transfer. HIGH
- **ISO 14971 risk-management plan integration** — linkedRiskItems[] should auto-pull from risk-register at PLANNING phase start. MEDIUM